TEAVISION360: A COMPLETE AI AND IOT PLATFORM FOR QUALITY, HEALTH, AND MARKET DECISIONS IN TEA FARMING.

Project ID: RP [25-26J-178]

Project Proposal Report

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Declaration

I hereby declare that the proposal I have submitted is entirely original to me and does not contain any content has already been submitted for a degree or diploma at another academic institution. I also declare that to the best of my knowledge and belief, it doesn't contain any material that has already been released or written by another person unless it is specifically acknowledged in the text. This declaration reaffirms my commitment to upholding academic honestly and integrity, and I accept full responsibility for the ideas in this proposal's originality and authenticity.

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The above candidate is conducting research for the undergraduate dissertation under my supervision.

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Abstract

In the global tea industry, accurate grading of processed tea leaves plays a vital role in determining product quality, market value, and customer satisfaction. Traditionally, tea grading in Sri Lanka has been performed manually by experienced tasters and graders, relying on their expertise to assess attributes such as leaf color, texture, size, and shape. While effective, this manual process is often subjective, time-consuming, and prone to inconsistency due to human error and fatigue. As the tea industry faces growing competition and increasing demand for standardized quality, there is a strong need for innovative, technology-driven solutions to enhance grading efficiency and reliability.

This research proposes the development of an **IoT-enabled AI-based tea grading platform** that leverages computer vision and deep learning techniques to automate the classification of tea leaves. High-resolution images of processed tea leaves are captured using a smart camera module integrated with environmental sensors that measure humidity and temperature—factors that significantly influence tea quality. The collected data is transmitted to a cloud platform, where a **Convolutional Neural Network (CNN)** model trained on local tea grades such as BOPF, OP, and Dust performs automatic grading. This multimodal approach ensures that both visual features and environmental conditions are considered, providing a more holistic and accurate evaluation.

The proposed system not only reduces manual effort but also ensures consistency, traceability, and real-time monitoring of grading outcomes. By integrating IoT and AI, the solution addresses a critical research gap in Sri Lanka's tea sector, where limited work has been done on combining smart sensing technologies with machine learning for automated tea classification. Furthermore, the platform can generate data-driven insights that support quality control, decision-making, and long-term improvements in tea production. Ultimately, this research aims to modernize tea grading practices, offering a low-cost, scalable, and efficient solution that strengthens the competitiveness of Sri Lanka's tea industry in the global market.

Keywords: *Tea Leaf Grading, Computer Vision, IoT, Convolutional Neural Networks, Image Processing, Smart Agriculture, AI in Tea Industry*

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1. Introduction

The most important agricultural export from Sri Lanka is tea, which contributes enormously to the economy of the national state and gives global recognition to Ceylon Tea. The quality and consistency of tea products, in fact, depend on their grading process, which involves the classification of tea leaves according to color, texture, shape, and size. Traditionally, tea grading in Sri Lanka has been carried out manually based on the expertise of tea tasters and graders. This assures that human intervention is done; however, it is often subjective, laborious, and prone to inconsistency.

The dynamic nature of the industry of tea and increasing global competition has made it imperative for tea grading to be more accurate, efficient, and standardized. Manual grading does not meet these requirements as human evaluation can often change because of different expertise, fatigue, or environmental conditions. Lack of automation, indeed, affects not only the grading consistency but also the productivity of quite large tea processing factories.

Recent advancements in technology, especially in Artificial Intelligence (AI), Computer Vision, and the Internet of Things (IoT), hold promising opportunities in improving this traditional practice. Automated classification of tea leaves can be achieved by delivering visual feature analysis methods such as shape, color, and texture through computer vision techniques. When combined with deep learning models such as convolutional neural networks (CNNs), the grading process can be made much superior in terms of precision and repeatability, compared to the traditional manual process.

More, this whole process can be enlarged by the introduction of the Internet of Things devices since they provide real-time data acquisition and monitoring. Smart camera modules can take high-resolution images of tea samples, whereas sensors can capture ambient temperature and humidity conditions because they directly influence tea quality. All this data is sent to cloud platforms for processing and enables the automation, standardization, and remote monitoring of grading while cutting down the dependency on human labor.

This research area seems to have a limited scope in Sri Lanka for the combined application of AI and IoT for tea grading, thus creating a significant gap for innovation. Most grading practices have been determined to be manual in practice, while very few studies explore computer vision approaches in conjunction with IoT for real-time grading and traceability. Addressing this gap, this research proposes the development of an AI-powered tea grading system enhanced with IoT technology, aimed at improving grading accuracy, efficiency, and scalability within local tea production.

Certainly, the study here aims at bringing modernization into the grading process through the dual application of AI and IoT, but also extending in terms of how the tea

sector in Sri Lanka reduces its dependence on the manual method, and thereby, adds possible value to the competitiveness of the Sri Lankan tea industry in the global world. Consistent data grading outcomes will go a long way in lessening subjectivity and introducing quality assurance into the system while continuing to support sustainable approaches in the tea sector.

1.1 Automating Tea Grading through Computer Vision and IoT Technologies

A traditional system for grading tea is influenced greatly by personalize observation which mostly shows subjectivity and in most cases inconsistency. Computer Vision and IoT will automate the grading system and enhance accuracy and efficiency. Computer Vision based on CNNs will classify the tea leaves depending on color, texture, shape, and size. The IoT devices such as smart cameras and environmental sensors fully build this system with real-time taking images from cameras and quality-related data (e.g., humidity and temperature); thus, creating reliability and scalability to be the best solution for tea grading in Sri Lanka.

1.1.1 Tea Leaves Classification using Images

Automated grading depends mainly on image processing, where high-resolution images of the processed tea leaves are taken with IoT-enabled smart camera modules before being sent to deep learning for analysis to extract the vital visual features needed. The CNNs will then train models on tea grades relevant to the country locally such as BOPF, OP, and Dust. This increases adaptability to the tea industry in Sri Lanka. By automation of this process, human bias in classification would be minimized and grading errors minimized and ensures that large-scale productions are consistent.

1.1.2 Contribution that IoT Provides in Real-Time Monitoring

The addition of IoT in tea grading is in the aspect of real-time collection program and transmission of information. While the major determinant for quality tea is harvesting, moisture, and temperature are environmental variables influenced by these variables as they alter the quality as it is exported. IoT-enabled sensors capture this data as well as images of tea to be sent to the cloud platform for further processing.

1.1.3 The Necessity for Automation in Tea Grading

Through automation, the three critical issues the tea industry faces will be combated. First, the teabags that will make grading independent of the human factor. Second, grading will take less time than the usual longer time-consuming processes. Third, machines are relatively consistent in grading, quite different than humans. Quality uniformity in batches graded through automated grading is insured. This technique cuts down the operating costs and creates scalability in production. Most importantly, data-driven automation makes the tea industry in Sri Lanka competitive by global standards where consistency and quality assurance are highly valued.

1.1.4 Method of AI and IoT Integration

IoT will be integrated with AI to include posterity in the approach by combining visual analysis and accompanying environmental monitoring. The first step involves capturing images of tea samples with IoT devices while recording environmental data. Then, these data are sent to cloud-based platforms where trained CNN models analyze and classify the images of tea leaves according to defined grades. Notably, environmental data add context for precisely assuring quality assessment. The standard grading output can come from this system, viewable via dashboards or mobile applications, thereby increasing transparency and efficiency within the tea factories.

2. Background And Literature Survey

Tea grading has one of the important impacts on the quality, marketability, and the world reputation that the tea industry enjoys for Sri Lanka. It has such national economic importance because it is one of the largest foreign currency earners for the country. Traditionally, the grading was only done by manual means by expert tea tasters who inspect tea leaves with their self-powered eyes and categorize them based on appearances such as color, texture, shape, and size into BOPF, OP, and Dust [1]. However, although this has been doing a great service for human expertise, it subjects such grading to a lot of subjectivity, inconsistency, and inefficiency particularly in a large-scale production environment [2].

Human judgment is one of the major hindrances in manual tea grading because it could change due to fatigue, experience level, and environmental conditions. This lack of uniformity in grading will affect the reliability of quality assurance very much both in local and export markets [3]. The process is also time-consuming and labor-intensive, thus creating higher operational costs and limiting scalability to meet increasing international demand. With this growing competition in the international tea market, Sri Lanka is under incessant pressure to modernize levels of tea grading yet preserving the fame of "Ceylon Tea" as a globally recognized brand [4].

Integrating new technologies, especially in Artificial Intelligence (AI), Computer Vision, and the Internet of Things (IoT), would provide a remedy to those effects. Computer vision, which encompasses deep learning models such as convolutional neural networks (CNNs), allows for classifying tea leaves with far greater precision and consistency-attained visual characteristics as compared to manual methods [5]. In addition to this, IoT devices take real-time pictures and generate quality-related data on environmental parameters such as humidity and temperature that directly affect tea quality which augurs well for integration with the current grading process [6]. Therefore, this will provide a data-driven view to the factories, guaranteeing higher accuracy, traceability, and efficiency in grading.

In the Sri Lankan perspective, poor research and implementation concerning AI and IoT in tea grading are visible. Limited studies have done image-based classification of tea leaves; however, very few have merged that with IoT-enabled environmental monitoring to develop an automated, comprehensive system [7]. These are great opportunities to design and implement low-cost, scalable solutions that would meet the unique needs of local tea factories. These systems would drastically reduce subjectivity, increase grading efficiency, and ensure uniform outcomes-thereby boosting the global competitiveness of Sri Lankan tea [8].

Thus, upgrading tea grading is a necessity and timely modernization to address the fundamental limitations of manual grading, leveraging emerging technologies to contribute to the development of the tea industry in Sri Lanka. The smart, automated grading solutions can provide a consistent quality of output, reduce operational inefficiencies, and corroborate the position of Sri Lanka in the international tea market.

2.1 Manual Tea grading Practices

The following literature review suggests key limitations of the existing technique of tea grading in the literature with especial focus on the tea industry of Sri Lanka. Typically, the grading of tea leaves is done manually with expert tasters by a human judgment of properties such as leaf color, texture, size, and shape [1]. Nevertheless, this method is subjective and unreliable because results are considerably affected by grader's expertise, fatigue, and environs [2]. As a result, the assignation by hand often differs across batches, the overall reliability of quality control is decreased [3].

Furthermore, manual grading is slow and labor-intensive which is not suitable for high-throughput production requiring speed and efficiency. One of the contributors to the productivity gaps in Sri Lanka's tea industry is the continued use of traditional manual methods [4]. The lack of uniform grading standards has aggravated the situation, as stated by Herath (2018), manmade procedures are not capable of ensuring the uniformity required in the international markets [3].

Another major drawback is the lack of traceability and data recording during manual grading. Even though human graders provide a result, without any digital record, it becomes nearly impossible to track quality trends or ensure accountability along the export chain [6]. This is a significant concern and highlights the urgent need for modernization by AI and IoT.

2.2 Computer version and AI tea grading quality control (Agriculture)

The recent advances in Computer Vision, coupled with the advances associated with Artificial Intelligence (AI) and Deep Learning models, especially CNN technology, were used widely to classify agricultural products like fruits, grains, and tea through visual features like, texture, and color [5][6]. The research results have shown that classification based on CNN results in a far greater precision and consistency than the conventional method of manual classification [7].

Table 1: Application of AI in agriculture product grading

Product	Technique used	Outcome	References
Tea Leaves	CNN- based image classification	Improved accuracy in grading [7]	Wijesinghe K (2022)
Coffee Beans	Image Processing + ML	Detection of defective beans [9]	Kumar et al. (2021)
Rice	Deep CNN Models	Grain type identification [10]	Li & Zhang (2020)
Fruits	Color & Texture Analysis (AI)	Ripeness detection [11]	Zhao et al. (2019)

2.3 Critical Analysis of work

This section looks at important factors that affect the development of automated tea grading solutions. These include image-based classification, IoT-enabled monitoring, standardization, and scalability.

2.3.1 Image based classification of Tea leaves

Recent research shows the promise of Computer Vision techniques, especially Convolutional Neural Networks (CNNs), for classifying agricultural products. Li & Zhang (2020) proved that deep learning models can accurately capture complex visual features like texture and shape. In Sri Lanka, Wijesinghe (2022) used CNNs to classify tea grades like BOPF and OP, achieving better accuracy than manual grading.

However, current implementations have limitations regarding dataset size, model generalization, and real-time adaptability. Most studies depend on small, lab-based datasets that do not reflect the variability of large-scale tea production.

2.3.2 Role of IoT in real time monitoring

IoT technologies add intelligence by capturing environmental factors like humidity and temperature. These factors directly impact tea quality. As a following Research paper Perera emphasized how IoT is changing agriculture, especially in continuous monitoring and data-driven decision-making [6]. In tea production, IoT-enabled smart cameras and sensors send real-time images and environmental data to cloud platforms for analysis.

However, current studies are mostly limited to proof-of-concept experiments. Once the investor who created an IoT monitoring system for greenhouses, but there are few similar large-scale IoT applications in tea grading [10].

2.3.3 Standardization and traceability in tea grading

Consistency and traceability are primordial in the international markets. In this respect, manual grading does not yield data records required by exporters and regulators. The automated system, on the other hand, can simply log grading histories that can support the claims of adherence to international quality standards [8]. Fernando who researched in 2019 claimed that an AI-integrated grading system could further boost Sri Lanka's international competitiveness, as AI could minimize subjectivity and ensuring repeatability of quality [1]. But the absence of harmonized datasets and standards of AI-based grading in Sri Lanka remains a major challenge.

2.4 Emerging treads in automated tea grading

The recent studies emphasize combining AI with IoT platforms for a systemwide grading perspective; they stress that being able to combine image recognition with sensor monitoring would allow consideration of visual attributes and environmental conditions [5][6]. Advanced CNN architectures are explored to improve classification accuracy for the studied agricultural products [9].

Thus, cloud-based IoT platforms are facilitating real-time analytics and remote monitoring, which extends grading capability beyond the factory floors [10]. Newer approaches, involving clustering algorithms for detecting latent quality features and reinforcement learning to continuously optimize classification models, indicate a shift toward more data-driven, standardized, and scalable approaches of tea grading.

3. Research Gap

While tea is one of the main exports of Sri Lanka, grading is still done mainly manually, with human judgment being the main basis for such systems. The traditional approaches have been criticized for subjectivity, inconsistency, lack of traceability, and inefficiency with large-scale production operations [2][3][4]. Some applications of computer vision deep learning have been noted in other agricultural practices such as fruit sorting and coffee bean classification [5][9], but in the case of Sri Lankan tea, studies have been limited to experimental or small-scale studies [7].

AI techniques such as Convolutional Neural Networks (CNNs), in studies pertaining to the current research on tea grading, have shown fairly good results in terms of accuracy and consistency [1][7]. However, these studies have been mostly lab-based and have been focusing on datasets, lacking scalability and industry-level validation. Most of these approaches also deal mainly with visual classification of tea leaves and ignore other important quality factors, such as humidity and temperature, that directly affect tea quality contaminants during processing and storage [6].

On the contrary, IoT research in agriculture has shown strong promise for real-time monitoring of environmental situations [6][10]. However, little to no incorporation of IoT with AI-based grading systems exists in tea production in Sri Lanka, thus further creating an apparent gap in AI and IoT combined into a complete solution that provides visual grading of the tea leaves and environmental factors affecting the quality of the tea.

Moreover, currently, the research is lacking in standard datasets and traceability mechanisms, which render difficult the evaluation and comparison of different techniques. The absence of these factors greatly limits the possible scaling and use of automated grading systems in the Sri Lankan tea factories.

Thus, research currently only opens a gap in the development of a cheap, integrated, and scalable AI + IoT automated tea grading system for Sri Lanka. Filling of the mentioned void will enable the industry to enhance the accuracy, consistency, and traceability of grading with reduced reliance on manual labor, thus ultimately strengthening global competitiveness of Ceylon Tea.

4. Research Problem

The study's research question is as follows: "Can an automated tea classification system powered by artificial intelligence and IoT be developed to address the following issues in the Sri Lankan tea industry?"

- **Manual Evaluation Is Subjective and Prone to Variability**

  The fact that manual ratings are subject to human processing constraints and, thus, tend to be inconsistent and highly variable due to the inherent nature of the method. Depending on the level of experience of the testers, their tiredness or just the environment they are in, there can be big differences in the outcome for the same batch of tea. Thus, the research question is how to automate the grading process to eliminate human bias and obtain uniform results.

- **Integrating Visual Elements and Environmental Factors**

  Most of the existing studies on AI tea grading have focused on grading the leaves based on image classification. Additionally, environmental factors such as humidity and temperature affect the quality of processed tea. The research problem might include how to integrate different computer vision models with data from IoT-enabled sensors as the grading becomes more comprehensive and accurate at this stage.

- **Scalability and Real-Time Implementation**

  Although numerous experimental studies have established the feasibility of CNNs for tea classification, these experiments are confined to a laboratory setting. Implementing such systems in real factories where large quantities of tea are graded poses certain challenges. The research question also ties back to the "accuracy and scalability" aspect since the system needs to be industry-level in terms of cost.

- **Standardization and Traceability**

  Since manual marking is not in electronic format, it may be difficult to ensure that there is accountability and follow-up for quality control. The global tea market calls for data-driven transparency and standardization. Therefore, the research question to increase the export competitiveness of Sri Lankan tea is how a system that ensures digital traceability and consistency in grading results can be developed.

The research question is how to create a verifiable, scalable integrated, and automated tea grading system in respect to AI and IoT. The study seeks to close the gap between the traditional and modern approaches by addressing subjectivity, combining visual and environmental factors, being scalable, and seeking standardization.

5. Objectives

5.1 Main objective

This research shall mainly focus on the formulation of a full-fledged automated AI and IoT-enabled tea grading system that could classify tea leaves with more precision according to their visual characteristics and their environmental factors, thereby reducing any subjectivity, improving consistency, and ensuring standardization of quality in Sri Lankan-produced tea.

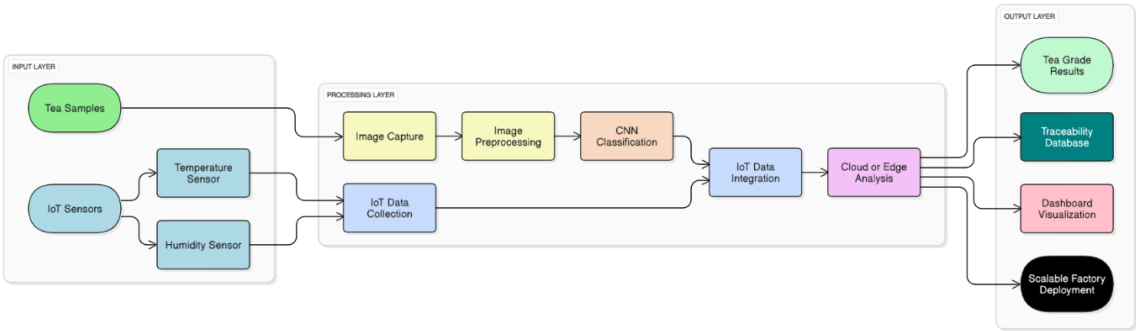


Figure 1: Research framework for proposed system

5.2 Specific Objectives

- **Pioneering Automated Image-Based Tea Leaf Classification**

The methodologies and deep learning engines (like CNNs) will be developed and engineered for the classification of tea leaves within an accepted grading standard as BOPF, OP, etc., while at the same time analyzing some of the major visual factors such as color, texture, shape, and size.

- **IoT Integration into the Environmental Monitoring System Grading Process**

Design and develop IoT devices such as temperature and humidity sensors during the grading process of the collection of real-time environmental conditions to obtain additional factors affecting quality.

- **Real-Time Data Transmission and Cloud Processing**

Develop a system of communication that allows IoT devices to send captured images and sensor readings toa cloud-based platform so that these will be processed in real time by the AI model and then will return grading results.

6. Methodology

6.1 Overall System Description

The proposed AI and IoT-Based Tea Leaf Grading System will be implemented using a microservices-based, modular architecture powered by state of-the-art computer vision and machine learning pipelines. The system adopts IoT enabled smart cameras and environment sensors and collects the images of tea leaves and the value of the humidity and temperature from the air automatically. These multimodal data inputs are forwarded to either a cloud or edge platform, in which a convolutional neural network (CNN) receives and determines, based on visual characteristics like color, texture, shape, and size, the grade of the tea. Combining IoT sensors with AI-based grading provides consistency, traceability, and scalability. Factory personnel and quality control inspectors can read results on-line on a scorecard, latest data also being in stored in Traceability Databases to support quality assurance and export compliance.

System workflow

As Fig. 1 illustrates, the entire cycle has been covered through system workflow, from data input-the tea-leaf samples, information from IoT sensors-to result output, involving graded classifications of teas, traceability records, and dashboard visualizations.

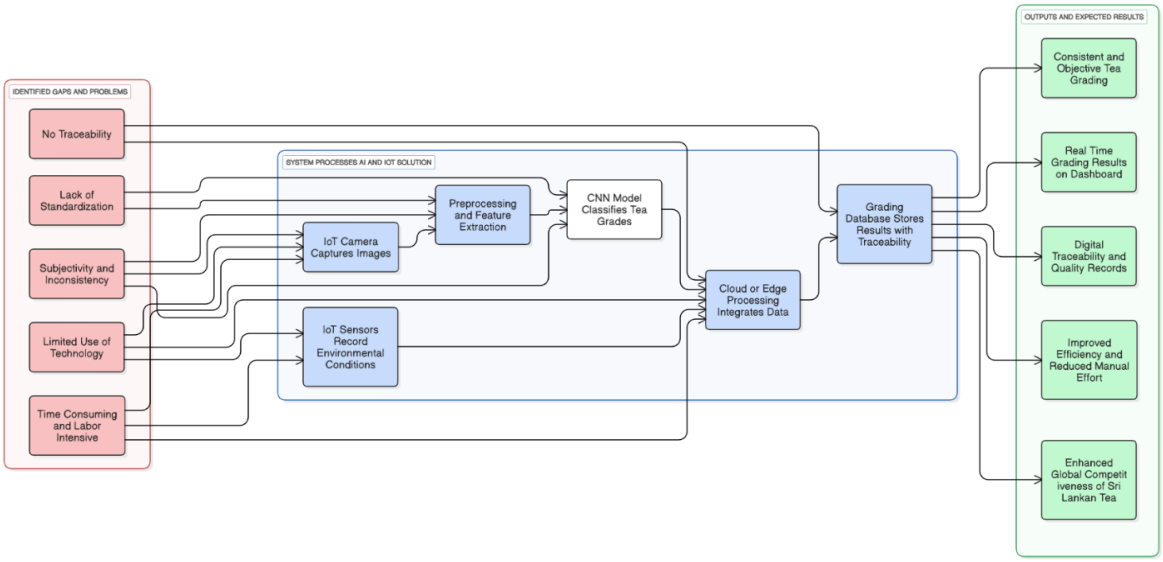


Figure 2: Conversational AI System Workflow Diagram

- **Input Layer** – The tea samples are kept under the IoT enabled smart camera model whereas the environmental (temperature and humidity) conditions are sensed by IoT sensors.
- **Processing Layer**

    - Image Capture and Preprocessing: The images of tea samples are captured in high resolution and preprocessed to improve feature extraction.
    - CNN Classification: A pre-trained CNN processes image features (color, texture, shape, size) and outputs the grade.
    - IoT Data Collection & Integration: Sensors (temperature, humidity) with image-based classification for enhanced grading analysis.
    - Cloud/Edge Analysis: The image data and sensor data are processed to give the precise grading to establish consistency and monitor in cancer real time.

- **Output Layer:**

    - Tea Grading Report: Auto generated and presented for operations.
    - Traceability Data Warehouse: Archives test results together with environmental information for enhanced quality control and regulatory compliance.
    - Dashboard: An interactive interface for the factory floor personal to be able to see grading results, trends and performance insights.
    - Factory Deployment Scale-Out: Ability to deploy across multiple production lines or factories.

6.2 Software Development Methodology

The development of an AI and IoT Based Tea Leaf Grading System would be based on the Agile Software Development Methodology, which would allow flexibility for the iterative improvements in machine learning model performance and IoT integration. The methodology emphasizes continuous development and testing, as well as stakeholder feedback, making sure that the system meets the practical requirements of the tea industry.

i. Requirements Evaluation

- Identify mechanically functional requirements such as the automated image capture, CNN-based grading, IoT sensor integration, cloud/edge processing, and dashboard visualization.
- Identify non-functional requirements such as the scalability of the system, low latency, cost-effectiveness, and accuracy of the grading results.

ii. System Design

- Design the modular architecture separating input using tea samples and IoT sensors, processing with image preprocessing, CNN classification, IoT data integration, and output: grading results, traceability database, and dashboards.
- Define communication protocol between IoT devices, cloud platforms, and database systems.

iii. Implementation

- Development of the camera module interface for automated image capturing. A separate IoT data collection module will be implemented to read temperature and humidity. The CNN model will then be trained on local tea grades (BOPF, OP, Dust, etc.) to gain higher accuracy. APIs will be developed to integrate the image classification results with IoT sensor data. The final step would be developing dashboard visualization tools for real-time monitoring.

iv. Testing & Validation

- Unit tests would be performed on individual modules (image preprocessing, CNN classification, and IoT data collection). Integration tests will verify the data flow across the modules is working smoothly. Validate results using expert manual grading to measure accuracy and reliability.

v. Deployment

- The deployment of the system either on the cloud or the edge server would keep it operational on the factory floor in real time. The dashboard will be integrated into the factory IT systems for operator accessibility. The system shall be expandable across multiple production lines.

vi. Iteration & Feedback

- On-going iterations to improve the accuracy of the CNN model using new datasets on tea samples. Input from users, e.g., factory operators and quality controllers, will be collected. The dashboard and traceability system will be revised to adapt to the ever-evolving industry.

6.3 Technology Considerations and Framework Selection

Selection of technologies for the AI and IoT-Based Tea Leaf Grading System will be based on performance parameters, commercial viability, ease of integration and scale. The system uses a mix of computer vision, IoT sensing, cloud/edge computing and visualization, and therefore, each of its components will require a framework/solution designed for its specific role.

Key considerations include:

- **Accuracy and scalability of AI models**: Specification of deep-learning frameworks, which support convolutional neural network (CNN) with GPU acceleration for efficient training and inference.
- **Compatibility for IoT devices**: The requisite framework should support lightweight communication and less power consumption with guaranteed data transfer capability.
- **Cloud and Edge computing capabilities**: The system must adopt scalable deployment and support real-time data processing.
- **Data storage and visualization**: The database should accommodate structured and unstructured data (images + sensor readings) while visualization frameworks should support interactive dashboards.

Table 2:Technology and Framework Selection

Component	Technology/Framework	Reason for Selection
AI/ Machine learning	Tensor Flow / PyTorch	Provide a strong support for the CNN- Based image classification.
Program Language	Python	Widely used in AI/ML, strong library support.
Image Processing	OpenCV	Efficient for preprocessing tasks for an example filtering and segmentation
IoT Device Platform	Raspberry Pi	Low cost, support real- time image capture and IoT sensor integration.
Sensors	DHT11 / DHT22	Reliable and low-cost temperature and humidity sensors to complement grading accuracy.
Communication Protocol	MQTT / HTTP REST API	Lightweight protocol (MQTT) for IoT device communication; REST API for cloud integration.
Cloud / Edge Platform	Firebase / AWS IoT	Provides scalability, real-time data processing, and secure cloud storage for images and sensor readings.
Database	PostgreSQL / Firebase Firestore	Supports structured data storage for sensor logs and grading results with scalability.
Dashboard Visualization	Grafana / Power BI / Custom Web Dashboard	Enables real-time monitoring, trend analysis, and traceability visualization for operators.

6.4 System Integration and API Layer

6.4.1 System Integration

- IoT Layer
  - The Raspberry Pi/ESP32-CAM will take images of tea leaves and record environmental parameters (temperature and humidity).
  - IoT communication with the cloud is realized through light-weight protocols like MQTT or HTTP REST.
- Processing Layer
  - OpenCV is used for preprocessing captured images (resizing, noise reduction, color normalization).
  - The CNN model will classify tea into various grades such as BOPF, OP, and Dust based on information learned from tea samples that are labeled.
  - Sensor readings are used in conjunction with image-based predictions to enhance grading accuracy.
- Cloud & Database Layer
  - Both processed results and raw data will be stored in a PostgreSQL/Firebase Fire store database for traceability.
  - The cloud/edge server manages and responds to AI inference requests from IoT devices through APIs.
- Visualization Layer
  - A web-based dashboard retrieves grading results and environmental logs through APIs.
  - With that, operators would be able to view real-time grading outcomes, historical trends, and environmental conditions.

6.4.2 Security Considerations

- **Authentication:** API keys or OAuth2 must ensure that unauthorized devices and dashboards do not have access to the services.
- **Encryption:** Communication is secured with HTTPS/TLS so that sensitive data can be protected.
- **Rate Limiting:** Calls to the API are observed to prevent misuse or overloading.

6.5 Testing And Validation Framework

Testing and verification processes ensure the reliability and accuracy of the AI and IoT-Based Tea Leaf Grading System, as well as its readiness for a real-world factory deployment. Elements of the framework include unit testing, integration testing, system validation, and performance evaluation with technical and user-centered approaches.

6.5.1 Testing Strategy

- **Unit Testing**

IoT Devices whether camera captures images correctly or not, and sensors (DHT11/DHT22) are accurate in recording temperature and humidity. The Preprocessing Module Resizing, color normalization, and noise reduction should be put through testing. The CNN Model should validate model loading, prediction function, and handling errors. The API Endpoints should verify whether each endpoint returns the correct response, for example, /upload/image, /classify/tea.

- **Integration Testing**

Check if any communication flaws exist in-between IoT devices, cloud server, database, and dashboard. Checking if images and sensor data flow into the pipeline without data loss. Testing if CNN results are in sync with environmental readings in the database.

- **System Testing**

Deploy the entire workflow in a controlled setup with test tea samples. Check grading outputs against grading by human experts. Monitor the dashboard for near-real-time updates to ascertain system responsiveness.

- **Validation Against Ground Truth**

The ground truth will be based on expert-graded datasets of tea. Compare the AI predictions against the experts' results to compute accuracy, precision, recall, and F1-score. Ensure the system's accuracy is over 90% acceptable for deployment in the industry.

- **Performance Testing**

Measure the time lapse from capture of the image to grading output. Stress tests the system whenever it loads several IoT devices to send images all at once. Assess scalability by simulating the deployment as in-full factory implementation.

- **User Acceptance Testing (UAT)**

Tea factory operators are involved to validate the usability of the dashboard and included quality controllers. Feedback can be gotten from the end users on grading clarity, dashboard readability, and integration with existing workflow. Refine system features based on user feedback.

6.5.2 Validation Metrics

Table 3: Validation Metrics

Category	Metric	Target
AI Model Accuracy	Precision, Recall, F1-Score, Confusion Matrix	≥ 90%
Sensor Accuracy	Error margin compared to calibrated tools	± 2%
System Latency	Time from image capture → result display	< 3 seconds
Data Reliability	Successful image + sensor data transfer rate	≥ 99%
Scalability	Maximum number of devices supported without failure	50+ concurrent IoT nodes
User Feedback	Operator satisfaction score (UAT survey)	≥ 80% positive

7. Project Requirements

7.1 Functional Requirements

The system must satisfy the following functional requirements concerning the grading of tea leaves with the aid of IoT and AI:

- **Image Capture**: The IoT camera module should take automated high-resolution images of tea samples.
- **Environmental Data Collection**: Sensors will measure temperature and humidity in real-time.
- **Transmission of Data:** IoT devices should securely transmit images and sensor data to the server/cloud (using MQTT/HTTP).
- **Image Processing & Classification:** A CNN model will image the teas and classify them into grades (e.g., BOPF, OP, Dust) that are already defined.
- **Data Storage:** Data concerning all images, sensors, and classification results needs to be centralized in a database for traceability.
- **Results Display**: The grading results should be displayed on a real-time dashboard, together with the data from the sensors.
- **Reporting:** The system shall generate reports that summarize grading performance over time.
- **API Support:** APIs will give external systems access to grading results, sensor logs, and dashboard data.

7.2 Non - Functional Requirements

- **Performance**: The entire grading process taking less than three seconds. as in image capture -- classification -- result display.
- **Scalability**: The system shall be fully operational with more than 50 IoT devices connected, without degrading their performance.
- **Reliability**: The transmission of data in the least shall be realized at above 99% success rate without any packet loss.
- **Accuracy**: The CNN model shall provide an accuracy of above 90% compared to expert grading.
- **Usability**: The dashboard shall be equipped with all kinds of user-friendly navigation and visualization.
- **Security**: All data exchanged shall be protected with encryption technology such as HTTPS or TLS.
- **Maintainability**: The system is supposed to allow easy updates of its CNN model and IoT firmware.

7.3 User Requirements

The system is meant for tea factory operators, quality controllers, and managers. Their requirements include:

- An easy-to-use interface for real-time viewing of tea grades.
- Monitoring of environmental conditions (temperature & humidity).
- Accessing historical grading results for quality audits.
- Manual efforts kept to a minimum compared to traditional grading.
- Mobile or web-based access gives flexibility to the dashboard.

7.4 System Requirements

Hardware Needs

- IoT Device: Raspberry Pi/ESP32-CAM with a camera module.
- Sensors: DHT11 / DHT22, which measures humidity and temperature.
- Server/Cloud: Atleast 4core CPU, 8GB RAM, with GPU support for CNN inference.
- Dashboard Access: Desktop PC, laptop, or tablet with internet connection availability.

Software Requirement

- High-level Language: Python (with OpenCV, TensorFlow/PyTorch).
- Database: PostgreSQL/Firebase Firestore.
- Communication Protocols used are MQTT, HTTP REST API.
- Dashboard: This might be created using Grafana. Alternatively, a web-based dashboard could be built using Flask or React.
- Version Control: Git/GitHub for collaborative development.

7.5 Expected Test Cases (Tentative)

Table 4: Expected test cases

Test Case	Description	Expected Result
TC1	Verify camera captures tea leaf image	Image successfully stored with timestamp & ID
TC2	Verify sensor readings from DHT11/DHT22	Accurate temperature and humidity values logged
TC3	Verify CNN classification	Tea grade (BOPF/OP/Dust) returned with confidence score
TC4	Validate API endpoint `/upload/image`	Image successfully uploaded → 200 OK response
TC5	Validate dashboard result display	Dashboard updates with correct tea grade in <3s
TC6	Validate integration of sensor + classification data	Result displayed with linked environmental readings
TC7	Test system with multiple IoT nodes (scalability)	System handles concurrent requests without failure

7.6 Wire Frames

Wireframes of the website and mobile application allow us to appreciate the entire user journey and interaction avenues that the tea leaf grading system lays out, from real-time tea grade results, environmental condition monitoring (temperature and humidity), to historical data tracking, all presided over by AI-powered image classification and IoT sensor integration. The web wireframes (Figures 3) present the Mobile wireframes showing Login, Dashboard, and Reports screens of the Tea Leaf Grading System. while mobile-responsive wireframes (Figures 7.4 and 7.5) demonstrate the simplified, on-the-go access to tea grade results and sensor data, allowing for usability across multiple devices.

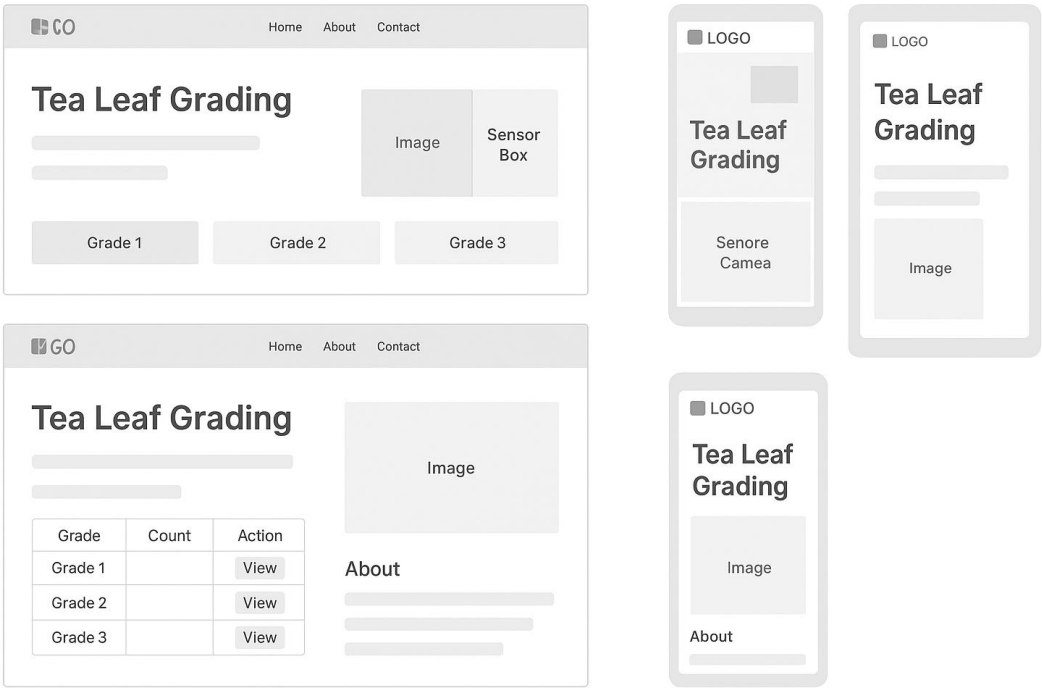


Figure 3: Mobile wireframes showing Login, Dashboard, and Reports screens of the Tea Leaf Grading System.

8. Budget And Budget Justification

The proposed budget caters for the essential hardware and software resources for the development and testing process of the automated tea grading system. The core IoT setup for image capture and processing consists of either a Raspberry Pi or ESP32 with a camera module. The sensors (DHT22 and odor sensor) provide vital environmental and aroma data necessary for grading accuracy. Power supply and cabling help ensure continuous operation of the device, and the cloud hosting allows real-time data storage and AI processing. Random costs will account for prototyping and testing expenses. Hence the overall estimated budget of ~40,000 LKR presents a cost-effective and yet fully functional solution for realizing the proposed research system.

Table 5 5: Budget and Budget Justification

Item	Quantity	Cost (LKR)	Purpose
Raspberry Pi / ESP32	1	15,000	IoT processing unit
Pi Camera / ESP32-CAM	1	8,000	Image capture
DHT22 Sensor	1	2,500	Temperature & humidity monitoring
Odor Sensor	1	3,500	Aroma detection
Power Supply & Cables	1	2,000	Device power
Cloud Hosting (Firebase/ThingSpeak)	-	5,000	Data storage & processing
Miscellaneous	-	4,000	Prototyping costs

9. Commercialization

The proposed IoT-based AI Tea Leaf Grading System is going to fill a very important gap for the tea industry in Sri Lanka, where grading is still done manually, making it inconsistent, subjective, and inefficient. This is a direct response to the needs that tea factories, exporters, and plantation owners have to increasingly automate, standardize, and sometimes provide real-time tea grading - in terms of wanting a more accurate, efficient, and traceable method of grading their tea. With all this in mind, it can be assessed that there is indeed potential demand that could make the proposal feasible because of Sri Lanka's need to maintain its strength and competitive edge in the global tea market as it adopts modern technologies. The idea is well supported for feasibility as IoT devices become cheaper, advanced AI image processing tools are being developed, and digital transformation in agricultural production is stepping up the push in the market.

- **Phase 1: Development and Pilot Testing of New Products**

  Develop a functional prototype for camera integration, IoT sensors, and CNN-based grading. Pilot deployments in local tea factories will be launched to show their accuracy and cost-effectiveness.

- **Phase 2: Freemium format for Small Producers**

  A low-or no-cost version of the platform (with image grading only) for small factories will be built. Other advanced features will be included in the premium subscription: sensor integration, cloud dashboards, and automatic reports.

- **Phase 3: Licensing to Large Tea Factories and Exporters**

  The system will then be customized to meet the enterprise-level needs of large factories, for instance, those related to batch grading, production analytics, and integration with ERP systems. Develop flexible licensing mechanisms and long-term partnerships.

- **Phase 4: Building into Scaling and Global Expansion**

  Tailoring the system to various international grading standards would reinforce India's, Kenya's, and China's tea markets. Such worldwide adoption will also be aided by strategic collaborations with the country's tea boards, exporters, and AgriTech companies.

10.Research Planning and Timeline

10.1 Planning

The actual research period can extend from 6-8 months, during which the research can be segmented into different phases. Every phase will have some significant deliverables that will enable the project or work to proceed systematically and within the expected times.

- **Phase I-Enactment Process on Project (Month 1):**
  Define the scope of the research, set the objectives, finalize the methodology, and establish an environment for hardware and software.

- **Phase II-Data Compilation (Month 2):**
  Collect and label images of tea leaves (BOPF, OP, Dust) and note readings from sensors (temperature, humidity, odor).

- **Phase 3-Model Design (Months 3-4):**
  Preprocess data sets, perform training on CNN model, tuning hyperparameters and environmental data.

- **Phase 4-Integrate IoT System (Month 5):**
  Connect Raspberry Pi/ESP32 integrated with camera and sensors and send data to the cloud platform.

- **Phase 5-Testing & Validation (Month 6):**
  Comparison of AI grading with grading made by human experts, accuracy evaluation, and model refinement.

- **Phase 6-Documentation & Reporting (Month 7-8):**
  Complete the research findings, technical documentations preparation, and submission of the final project report.

10.2 Work Breakdown Structure

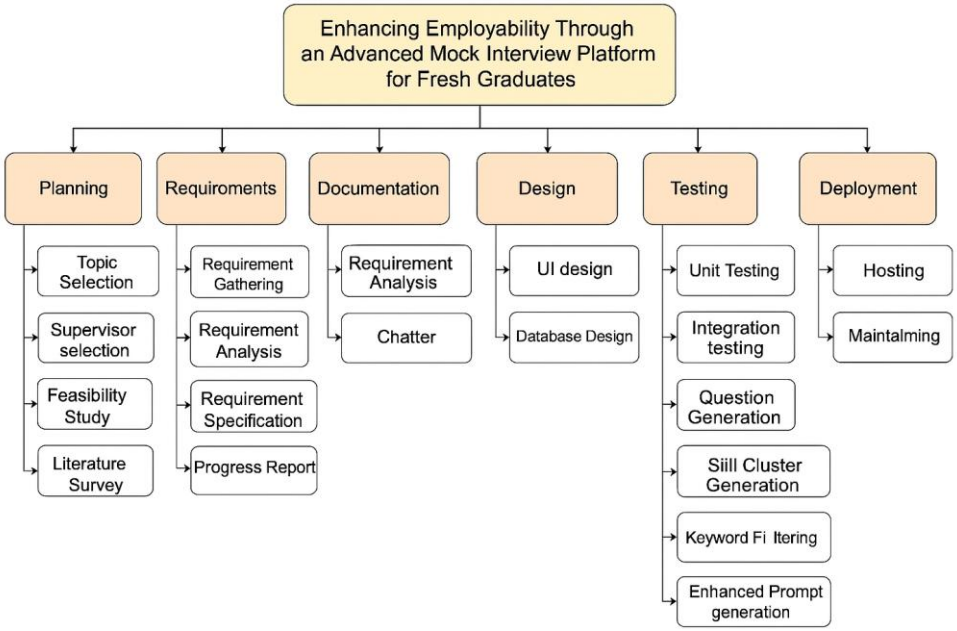
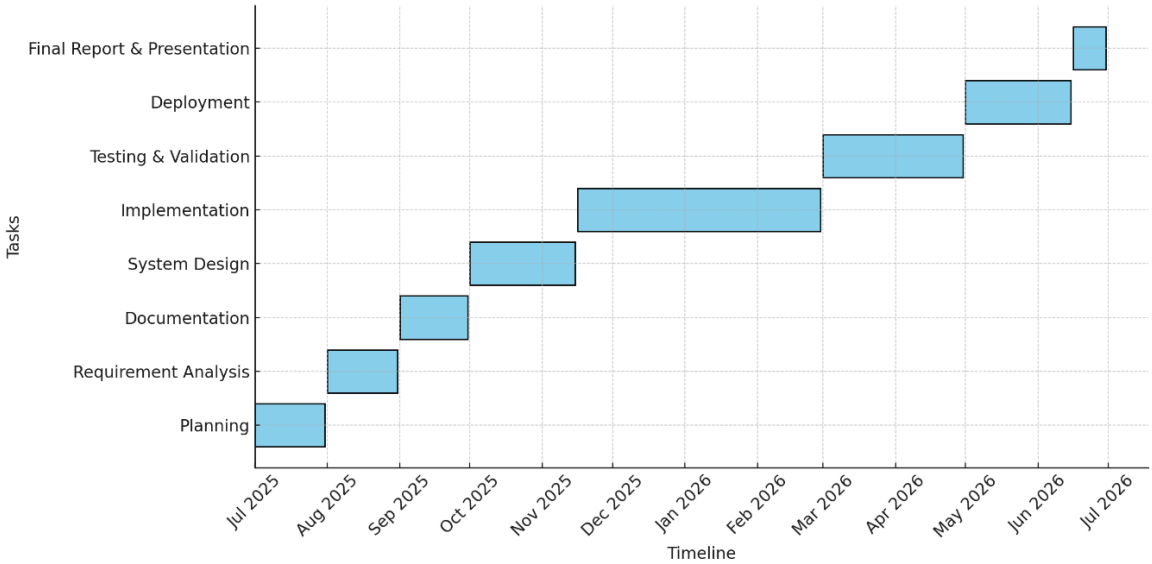


Figure 5: Work breakdown structure

10.3 Gantt Chart



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